TAL time-field mass and (λ, α_Y) exclusion: published Eöt-Wash bounds and the combined window (r3)

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Abstract

We translate published Yukawa-strength limits into a conditional mass constraint for the TAL time field, without fitting the experimental source model. The primary input is $|\alpha_{95}|$ in Lee's thesis Table 11.4 for Eöt-Wash 2020, with log–log interpolation between rows. For $\alpha_Y = 1$, the crossing is $\lambda = 98.1355\,\mu$m, or $mc^2 = 2.01076\,$meV. Within the tabulated domain, masses from $0.0221716\,$meV up to (but not including) this crossing mass are excluded; higher masses up to $39.46540\,$meV are not excluded. The sign-specific $+\alpha_{95}$ edge is $101.0286\,\mu$m and the digitized Fig. 5 edge is $98.0737\,\mu$m. The combined published envelope retains the micron-window edge $75.8771\,\mu$m ($2.60061\,$meV), set by HUST 2020, not Eöt-Wash. The rebuilt reduced source model did not pass genuine validation, so no TAL prediction is derived from that fit. This result is an external application of published experimental bounds, not an independent reproduction of their analysis. "Not excluded" is not confirmation.

1 Scope and conventions

We ask which masses and couplings survive a specified published bound. No best-fitting mass, force, or torque is inferred. We assume a universal, composition-independent, unscreened Yukawa addition to Newtonian gravity:

$$V(r) = -\frac{Gm_1m_2}{r}\left(1 + \alpha e^{-r/\lambda}\right), \quad \alpha = \frac{\alpha_Y}{4\pi}, \quad \lambda = L, \quad mc^2 = \frac{\hbar c}{\lambda}. \tag{1}$$

The fixed dictionary uses $\hbar c = 1.97326980 \times 10^{-7}\,$eV m (CODATA 2018 [5]). Thus $\alpha_Y = 1$ means $\alpha_{\mathrm{TAL}} = 1/(4\pi) = 0.07957747$, not $\alpha = 1$.

For a published bound $\alpha_{\mathrm{lim}}(\lambda)$ we define

$$\alpha_{Y,\max} = 4\pi\alpha_{\mathrm{lim}}, \qquad R = \frac{\alpha_{\mathrm{TAL}}}{\alpha_{\mathrm{lim}}}, \qquad F = \frac{1}{R}. \tag{2}$$

$R > 1$ is excluded (category b); $R \le 1$ is not excluded. R is a conditional ratio, never a sigma. F is a factor relative to a bound, not detector sensitivity. Outside an input's domain the result is not computed, not extrapolated.

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2 Published inputs and the calculation

The primary input is the published Table 11.4 $|\alpha_{95}|$ column [2], not a result of our source-model fit. Its 66 rows span 5 μm to 8.9 mm. We interpolate $\log \alpha_{\rm lim}$ linearly against $\log \lambda$ between adjacent rows only. The primary crossing lies between $(89\,\mu\text{m}, 0.097)$ and $(100\,\mu\text{m}, 0.0766)$. No α^* is interpolated.

The thesis defines two different bound constructions: $+\alpha_{95}$ and $-\alpha_{95}$ come from the $\Delta\chi^2 = 6.17$ surface; $|\alpha_{95}|$ is obtained by enclosing 95% of $N(\alpha^*, \sigma_\alpha)$ between symmetric endpoints. We do not interchange these columns. We report $+\alpha_{95}$ as the positive-coupling variant. The digitized Eöt-Wash 2020 curve in PRL Fig. 5 [1] is a cross-check, not the primary limit.

The combined envelope is the minimum of 17 packaged digitized curves from Lee Fig. 5 and Adelberger–Heckel–Nelson Fig. 5 [4]. It is contextual: it is not refitted, and it is not a newly combined statistical confidence region. Applying its drawn absolute-strength curves to positive α is an explicit reading assumption, particularly for HUST [3].

Recovered-file provenance

The original r4 archive is absent from this box. Three recovered member files have exactly the previously recorded SHA-256 values: the Table 11.4 CSV, the Eöt-Wash Fig. 5 CSV, and `external_bound/inputs/envelopes.npz`. Full digests and byte counts are shipped in `external_bound/input_manifest.json`. No claim is made that the absent archive itself was verified here.

The recovered `external_bound/inputs/results.json` is the r3 window output (SHA prefix `aa258c9577f53e1f`), not the absent r4 output. Its first excluded grid node is not the continuous crossing. We recompute the crossing from the verified envelope; `external_bound/provenance.json` records this distinction.

Preflight: passed

- All recovered inputs match their recorded digests. The 17 raw curves reproduce the covered envelope exactly (maximum relative deviation 0).
- The 23 Table 11.4/Fig. 5 F values agree with the recovered window table to a maximum relative deviation 6.4393×10^{-15} (tolerance 10^{-9}); the earlier recorded comparison was 0, so this difference is floating-point roundoff.
- All four crossings match the lost calculation's recorded values within 0.1%; the continuous envelope edge also matches the packaged window crossing to machine precision.
- Finite, positive bounds and strictly increasing range grids are required. Uncovered envelope cells are checked against the union of the 17 input domains; no NaN or Inf is accepted inside coverage. No missing key is ignored.

Evidence: `external_bound/preflight.json`, `external_bound/inputs/window_key_table.json`, and `external_bound/inputs/window_crossings.json`.

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3 Eöt-Wash 2020: results for $\alpha_Y = 1$

Published bound	crossing λ [μm]	mc^2 [meV]
Table 11.4 $\lvert\alpha_{95}\rvert$ (primary)	98.1355	2.01076
Table 11.4 $+\alpha_{95}$	101.0286	1.95318
Fig. 5 Eöt-Wash (digitized)	98.0737	2.01203
Combined envelope (context)	75.8771	2.60061

The primary and Fig. 5 limits agree at digitization level: Fig. 5/table ranges from 0.99615 to 1.00528 over 45 overlapping table rows. The two crossings differ by about 0.062 μm. This is a cross-check of two representations of the same experimental result, not independent evidence.

Key table

All bound columns below use the published Table 11.4 $\lvert\alpha_{95}\rvert$, interpolated where needed. At the crossing $R = F = 1$ by definition. See `external_bound/external_bound_key_table.csv`.

λ [μm]	mc^2 [meV]	$\lvert\alpha_{95}\rvert$	$\alpha_{Y,\max}$	R	F
5	39.465	8.45×10^{6}	1.0619×10^{8}	9.4175×10^{-9}	1.0619×10^{8}
10	19.733	3090	38830	2.5753×10^{-5}	38830
20	9.8663	21	263.89	0.0037894	263.89
40	4.9332	0.877	11.021	0.090738	11.021
50	3.9465	0.419	5.2653	0.18992	5.2653
63	3.1322	0.218	2.7395	0.36503	2.7395
75.877	2.6006	0.13855	1.741	0.57437	1.741
89	2.2172	0.097	1.2189	0.82039	1.2189
98.136	2.0108	0.079577	1	1	1
100	1.9733	0.0766	0.96258	1.0389	0.96258
110	1.7939	0.0641	0.8055	1.2415	0.8055
250	0.78931	0.029	0.36442	2.7441	0.36442
1000	0.19733	0.0192	0.24127	4.1447	0.24127
8900	0.022172	0.0116	0.14577	6.8601	0.14577

Mass intervals within the table domain

Excluded: $98.1355\,\mu\text{m} < \lambda \le 8.9\,\text{mm}$, or $2.217\times10^{-5}\,\text{eV} \le mc^2 < 2.0108\,\text{meV}$.

Not excluded (not confirmation): $5\,\mu\text{m} \le \lambda \le 98.1355\,\mu\text{m}$, or $2.0108\,\text{meV} \le mc^2 \le 39.47\,\text{meV}$. The exact crossing belongs to $R \le 1$.

Not computed by this source: $\lambda < 5\,\mu\text{m}$ or $\lambda > 8.9\,\text{mm}$. At 50 μm, $F = 5.26531$. At 100 μm the primary bound excludes $\alpha_Y = 1$, whereas $+\alpha_{95} = 0.0813$ gives $R = 0.9788$, still not excluded.

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4 The (λ, α_Y) diagram and the combined window

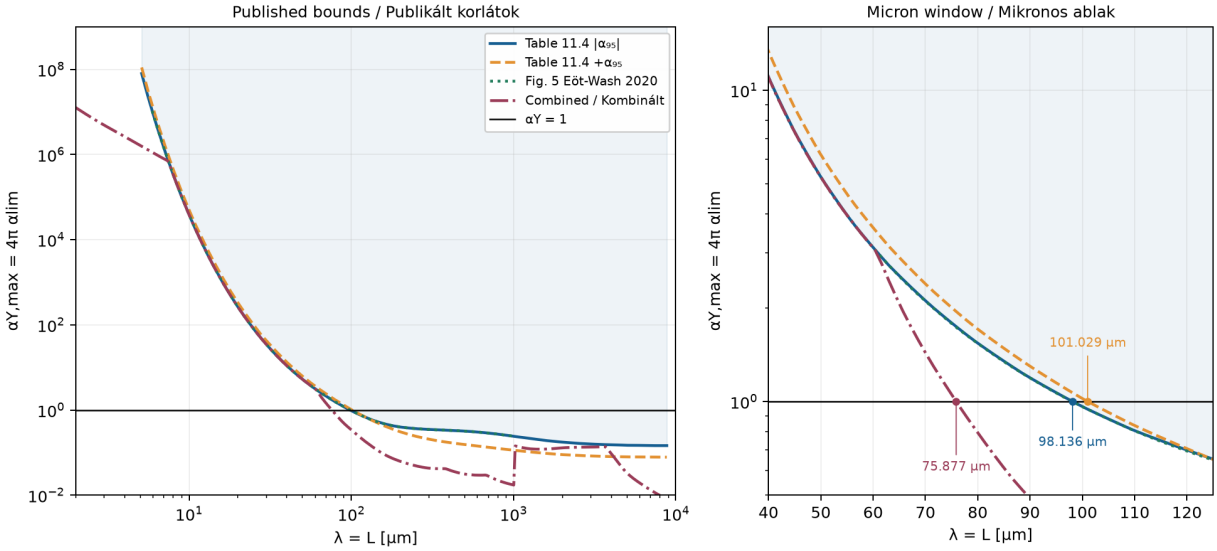


Figure 1: Published $\alpha_{Y,\max} = 4\pi\alpha_{\lim}$ versus the $\alpha_Y = 1$ branch; right panel: micron-scale detail. Pale shading above the primary curve marks exclusion by that curve. The combined envelope is context, not an independent statistical combination. Data: `external_bound/eotwash_external_scan.csv`; plotting: `external_bound/make_figure.py`.

The combined window is unchanged: over the specified search range, it is not excluded only for

$$2\,\mu\text{m} \le \lambda < 75.877\,\mu\text{m}, \qquad 2.6006\,\text{meV} < mc^2 \le 98.66\,\text{meV}. \tag{3}$$

HUST 2020, not Eöt-Wash, sets the large-λ edge. HUST closes the strip between about 75.877 μm and 98.136 μm (masses about 2.01–2.60 meV) which Eöt-Wash alone does not exclude. The packaged boundary convention uses an open upper end for the combined window; at the exact continuous crossing $R = 1$, so equality itself is not a strict $R > 1$ exclusion.

This statement is limited to the covered range (2 μm–1 cm in the scan). It does not establish an allowed interval at shorter ranges or make a new long-range claim. One short-range input is an unattributed drawn curve; the combined diagram preserves that provenance limitation. The envelope's step near 1 mm is the end of a digitized curve's coverage, not a new physical feature.

The experiment-specific and envelope statements must not be swapped: the Eöt-Wash edge is 98.1355 μm; the combined-envelope edge is 75.8771 μm. Neither is a measurement of the time-field mass.

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5 Source-model status, independence, and limitations

The rebuilt Eöt-Wash reduced source-model validation did not pass. The prior M2 variants with the documented inputs yielded no genuine validation. Formal threshold crossings were not accepted as validation. Consequently, **there is no TAL prediction from our source-model fit** and none of its fitted limits enters this article.

The validation criteria remain ±20% at at least 2 of the 7 specified points, with $\Delta\chi^2_{95} = 6.17$. We neither loosen them nor invent nuisance parameters. The 9.2 μm screen-offset hypothesis is not treated as a documented fact. Missing diagnostic artifacts were not reconstructed or rerun here; the negative status is an explicitly recorded prior result in `external_bound/provenance.json`, not a claim of a new source-model calculation.

What this external result is worth

It gives a reproducible dictionary conversion and a conditional exclusion for a specified coupling and force law. It is useful even without our source fit because the experimental limit is published. It is, however, **less independent** than reproducing the experiment's analysis: it inherits the authors' geometry, calibration, nuisance treatment, and systematics. Agreement between the table and figure checks transcription and digitization, not those underlying assumptions.

Caveats

1. The PRL Supplemental Material per-run position table was not retrieved. Its absence does not prevent use of the published bound, but prevents an independent source-model reconstruction. The unexplained roughly 8 μm separation offset in the prior diagnostics remains unresolved.

2. The $|\alpha_{95}|$ integral bound and the $+\alpha_{95}$ profile boundary are different constructions. The latter moves the edge from 98.14 μm to 101.03 μm; this difference is not hidden.

3. Log–log interpolation is an approximation between published table rows. The displayed digits reproduce the calculation, not an experimental precision claim.

4. The figure is digitized and the combined envelope carries graphical and attribution uncertainties. No new uncertainty model or tolerance is fitted here. HUST's drawn absolute-strength curve is used as packaged; no sign-separated HUST analysis is inferred.

5. Only the universal, unscreened Yukawa mapping is tested. Environment-dependent screening or alternative couplings require a different calculation.

6. R and F are conditional bound ratios, not significances or discovery probabilities. "Not excluded" is not confirmation, and no force or torque detectability calculation is supplied.

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6 Data, code, and reproduction

`TAL_time_field_mass_data_r3.zip` includes `external_bound/`, `article_source/`, `README.md`, `LICENSE.txt`, `requirements.txt`, `run_all.sh`, and `SHA256SUMS`. No third-party publication PDF is redistributed.

The primary calculation is `external_bound/external_bound.py`. It verifies input SHA-256, recomputes crossings directly from the table/curves/envelope, checks the recorded results and 23 window F values, and writes `external_bound_results.json`, `external_bound_key_table.csv`, `eotwash_external_scan.csv`, and `preflight.json` inside `external_bound/`. The package includes every local file cited here. The raw input digests are:

Recovered input	SHA-256 prefix
Table 11.4 CSV	`cf711b4fe52db194`
Eöt-Wash Fig. 5 CSV	`093fabd78879a17d`
`envelopes.npz`	`54eb95655554be45`
r3 `results.json`	`aa258c9577f53e1f`

Full hashes, byte counts and the absent-r4-archive provenance are in `external_bound/input_manifest.json` and `external_bound/provenance.json`.

Reproduction. In the extracted data directory, install the versions listed in `requirements.txt`, then run `./run_all.sh`. The bound script stops on a missing key, a non-finite covered input, a digest mismatch, or a crossing difference above 0.1% from the recorded result. The figure script reads the scan, and `article_source/build_articles.py` reads the JSON outputs to regenerate both articles. PDF rendering uses pdfLaTeX and the shared `article_source/tex/common.tex`. Sources contain no own source-model fit. The packaged results can be inspected without running any calculation.

Numerical cross-checks with Academia Co-scientist (AI-assisted); not peer review.

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References

[1] J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer, B. R. Heckel, New test of the gravitational $1/r^2$ law at separations down to 52 μm, Phys. Rev. Lett. 124, 101101 (2020), `https://doi.org/10.1103/PhysRevLett.124.101101`.

[2] J. G. Lee, A Fourier-Bessel Test of the Gravitational Inverse-Square Law, PhD thesis, University of Washington (2020), Table 11.4, `https://hdl.handle.net/1773/45542`.

[3] W.-H. Tan et al., Improvement for testing the gravitational inverse-square law at the submillimeter range, Phys. Rev. Lett. 124, 051301 (2020), `https://doi.org/10.1103/PhysRevLett.124.051301`.

[4] E. G. Adelberger, B. R. Heckel, A. E. Nelson, Tests of the gravitational inverse-square law, Annu. Rev. Nucl. Part. Sci. 53, 77–121 (2003), `https://doi.org/10.1146/annurev.nucl.53.041002.110503`.

[5] E. Tiesinga et al., CODATA recommended values of the fundamental physical constants: 2018, Rev. Mod. Phys. 93, 025010 (2021), `https://doi.org/10.1103/RevModPhys.93.025010`.